

A Field Guide to the WA Electricity Market Data

What every dataset in this project actually measures, why it looks the way it does, and the handful of concepts you need before you trust a single number out of it — explained with one running analogy, used throughout: the grid as a river, and batteries as water towers.

6

DATASETS COVERED

3

YEARS OF HISTORY

1

ANALOGY, USED THROUGHOUT

0

PRIOR KNOWLEDGE ASSUMED

Contents

01	Where this data comes from	— AEMO, the WEM, why WA is its own island, and the river analogy used throughout
02	Two units you must never confuse	— MW vs MWh, as flow rate vs volume
03	Two clocks running at once	— dispatch vs trading intervals, and AWST
04	The six datasets, one by one	
4.1	Operational demand & withdrawal	— how much water the river must supply
4.2	Distributed PV	— the rain barrels the grid can't see directly
4.3	Reference trading price	— what a bucket of electricity cost, every 30 minutes
4.4	Facility SCADA	— minute-by-minute water-tower behaviour
4.5	Facility registry	— the "who's who" of reservoirs and towers
4.6	Facility temperature	— the weather variable behind demand spikes
05	How the six datasets connect	— one diagram tying it together
06	Reading the raw files without getting burned	— five habits of the file format
07	Glossary	— every term, and the analogy key, in one place

Where this data comes from

Every file in this project comes from the **Australian Energy Market Operator (AEMO)**, the organisation that runs Australia's electricity markets. AEMO doesn't own power stations or wires — it's the referee. Every five minutes it looks at how much electricity people want, tells generators how much to produce, and publishes the price that resulted. All of that gets logged and published as open data, which is what this project reads.

DEFINITION

WEM (Wholesale Electricity Market) is the market AEMO runs specifically for Western Australia. **SWIS (South West Interconnected System)** is the physical grid it operates on — the network of poles, wires, and power stations connecting Perth, the South West, and the Wheatbelt.

The important thing that makes WA data simpler to reason about than most other regions:

WA's grid is an electrical island

Think of the SWIS as one river system: reservoirs and wells (power stations, wind and solar farms) feed a shared river, and every tap along the bank draws from that same flow. Unlike the eastern states — where New South Wales, Victoria, Queensland, and South Australia are all wired together into the National Electricity Market (NEM) and can ship power to each other when one state runs short — the SWIS has **no transmission link to anywhere else**. It is thousands of kilometres from the nearest other grid, across mostly empty desert. There is no pipeline in from a neighbouring river system. Whatever electricity WA uses, WA (and only WA) has to generate at that exact moment. This is why battery storage and demand-side flexibility matter so much in this data: there's nowhere else for the balancing act to happen.

WA doesn't observe daylight saving

Western Australia stopped observing daylight saving in 2009. That's a small political fact with a big practical consequence for this project: every timestamp in every file is a clean, fixed **UTC+8 (Australian Western Standard Time, AWST)** offset, all year round. There's no "the clocks jumped forward an hour" gap in spring and no duplicated hour in autumn — a problem that trips up almost every other Australian or US dataset. One less thing to debug.

WHY IT MATTERS FOR THIS PROJECT

Because the river is isolated, supply and demand have to balance locally, in real time, using only what's connected to the SWIS: gas and coal reservoirs, wind and solar wells, private rain barrels on houses (rooftop solar), and — increasingly — water towers beside the river (grid-scale batteries) that store surplus flow for later. That's exactly the cast of characters the six datasets in this guide describe.

A river runs through this guide

One analogy recurs in every section below, so it's worth stating once, in full, rather than re-explaining it each time. It isn't perfect — real electricity can't sit in a pipe the way water can, and grid frequency has to stay balanced within milliseconds, far tighter than any water network ever needs to — but for building intuition about flow versus volume, and about why a battery both draws from and feeds back into supply, it holds up well.

THE ANALOGY, STATED ONCE

WA ELECTRICITY CONCEPT	RIVER & WATER-TOWER EQUIVALENT
The grid (SWIS)	A river and pipe network delivering water from reservoirs to every tap, in real time
AEMO	The water authority, checking flow every five minutes and adjusting valves
Generators	Reservoirs and wells feeding the river
Power (MW)	Flow rate — what a gauge reads right now, in litres per minute
Energy (MWh)	Volume — how much water actually passed by, over some stretch of time
Grid-scale battery	A water tower beside the river: fills (charges) when there's a surplus, drains (discharges) to help supply when there's a shortage
Rooftop solar	A private rain barrel behind each house, feeding that house's own taps directly — invisible to the water authority
Operational Demand	Total flow the reservoirs must supply, including water diverted to refill the towers
Price (\$/MWh)	The going rate per bucket at the reservoir gate, that half-hour

SECTION 02

Two units you must never confuse

Almost every mistake a beginner makes with this data traces back to mixing up two units that sound similar but measure completely different things: **megawatts (MW)** and **megawatt-hours (MWh)**.

DEFINITION

Power (MW) is a *rate* — how fast energy is flowing right now.

Energy (MWh) is an *amount* — how much energy flowed in total over some period of time.

Picture a gauge mounted on the pipe leaving a reservoir. It reads a flow rate right now — how many litres are moving past this instant. Power (MW) is that gauge. Now picture a tank that collects whatever passes through the pipe over some stretch of time. Its level reads a total, accumulated amount. Energy (MWh) is that tank. You cannot read one off the other without also knowing how long the pipe ran for.

```
Energy (MWh) = Power (MW) × Time (hours) // e.g. a 100 MW generator running flat out for
30 minutes (0.5 h) produces 50 MWh
```

WHERE THIS BITES IN THE ACTUAL DATA

The battery SCADA files (Section 4.4) report **Average MWh** — energy delivered over a 5-minute dispatch interval, *not* a power reading. It's a tank-level reading ("this much moved past in this one check"), not a gauge reading. To compare it with every other series in this project, which are all in MW, you have to convert: since 5 minutes is 1/12 of an hour, multiply the MWh figure by 12 to get average MW. The project's loader (`load_facility_scada`) does this for you automatically and hands back an explicit `mw` column — but it's worth knowing why that conversion exists, because plenty of public AEMO tables don't do it for you.

SECTION 03

Two clocks running at once

This project mixes data recorded on two different clocks, and understanding the difference matters for anything you aggregate or compare.

CLOCK	LENGTH	USED BY	WHAT "ONE ROW" MEANS
Dispatch interval	5 minutes	Demand & withdrawal, distributed PV, battery SCADA	AEMO's physical control loop — the river is actually re-checked and re-valved this often
Trading interval	30 minutes	Reference trading price	The financial settlement window — the price generators actually get paid

Think of the dispatch interval as how often the water authority walks the length of the river and adjusts every valve — every five minutes, in this analogy. The trading interval is the billing window: six of those valve-checks bundled together into one 30-minute invoice for "what water cost during this half hour." This project's helper function `to_trading_intervals()` exists specifically to average the 5-minute series up to 30 minutes so they can be compared against price on equal footing.

LABELLING CONVENTION

AEMO labels a trading interval by its *start* time. The row stamped `12:00` covers everything from 12:00 up to (but not including) 12:30 — i.e. the six dispatch intervals at 12:00, 12:05, 12:10, 12:15, 12:20 and 12:25. It is easy to assume a timestamp marks the *end* of a period instead; AEMO does the opposite.

All timestamps are local Perth time (AWST)

Every timestamp column in every file — demand, price, PV, SCADA, temperature — is already in AWST (UTC+8), and (per Section 1) that offset never changes across the year. You do not need to do any timezone conversion to compare series against each other. You do, however, need to know that the underlying *files* are cut on UTC day/year boundaries rather than local ones — see Section 6.

SECTION 04

The six datasets, one by one

Each of the following is one AEMO data product this project loads and joins together. For each one: what it physically measures, the concept you need to interpret it correctly, and the single most common way to misread it — tied back, where useful, to the river-and-water-tower analogy from Section 1.

4.1 Operational Demand & Withdrawal

OperationalDemandWithdrawal-{2023..2026}.csv

Resolution: **5 min**

Unit: **MW**

Rows loaded: **307,008**

This is the closest thing to "how much electricity is Western Australia using right now" — in river terms, how much flow the reservoirs must put into the river this instant. But the file actually contains *three* related demand numbers, not one, and the difference between them is one of the most useful facts in the whole project.

THE THREE COLUMNS, DEFINED

Operational Demand — the total power that grid-connected generators are sending out to meet, including transmission and distribution losses along the way. This is the total flow the reservoirs must put into the shared river, net of rooftop solar: if your house fills its own rain barrel and uses that water, it never enters the river and never shows up here.

Unscheduled Operational Demand — Operational Demand with one more thing subtracted: the water being diverted by the towers (battery storage) as they refill. This is the closest number to genuine, underlying *tap* demand — homes, factories, offices, shops.

Operational Withdrawal — the flow the towers are drawing from the river to refill. Always zero or negative in this file (range: -1,158 to -0.40 MW), because "withdrawal" is measured as a subtraction from supply.

These three aren't independent — this project's data-quality audit verified the relationship holds exactly (to rounding error) across all 307,008 five-minute intervals in the dataset:

```
Operational Demand = Unscheduled Operational Demand - Operational Withdrawal //
withdrawal is ≤ 0, so this is really: Operational Demand = Unscheduled Demand + tower refill
```

Unscheduled Operational Demand
genuine tap demand

+

|Operational Withdrawal|
towers refilling

=

Operational Demand
what the reservoirs must supply

WORKED EXAMPLE

Suppose at 2pm, Unscheduled Operational Demand (genuine tap use across WA) is 2,000 MW, and the state's water towers are collectively refilling at a combined 150 MW. Operational Withdrawal is recorded as -150 MW. Operational Demand — the figure the reservoirs actually have to supply into the river — is $2,000 - (-150) = \mathbf{2,150 \text{ MW}}$. The towers added 150 MW of extra flow on top of what people were actually using, exactly the way a water utility has to pump extra volume to refill its towers on top of the town's ordinary tap demand.

WHY IT MATTERS

Because they now move in *opposite* directions at times, "minimum demand" — a headline grid-planning statistic — means something different depending on which of the two demand series you quote. Operational Demand is flattered upward by the towers refilling; Unscheduled Operational Demand strips that back out to show what taps alone are doing. A responsible analysis reports both.

4.2 Distributed PV (Rooftop Solar)

distributed-pv-new-{2023..2026}.csv

Resolution: **5 min**

Unit: **MW**

Rows loaded: **307,007**

Not every solar panel in WA is a big, purpose-built solar farm that AEMO can watch directly. Hundreds of thousands of houses have their own small rain barrel — a rooftop panel system that fills and empties locally, behind that household's own tap. The water is collected and used (or overflowed back) right there, invisible to the water authority, from the grid operator's point of view.

DEFINITION

Behind-the-meter generation is electricity produced on the customer's side of their meter — rooftop solar being the classic example — as opposed to **grid-scale** generation from a registered power station that AEMO dispatches directly. Because AEMO cannot see behind-the-meter generation directly, this file is an **estimate**, modelled from things like weather, known installed rooftop capacity, and time of day/year — not a metered reading.

This is the missing piece that connects to Section 4.1: Operational Demand is *net* of rooftop solar, meaning the rain-barrel water has already been subtracted out before that file was even published. Distributed PV is how you get it back, if you want to reconstruct *gross* underlying consumption — what people would be drawing from the shared river if nobody had a rain barrel at all.

COMMON PITFALL

It's tempting to treat Distributed PV like any other generator's output. But because it's estimated rather than metered — nobody can put a gauge on every rain barrel in the state — and because it's the reason WA's demand curve dips so dramatically around midday (the so-called "duck curve" effect, where grid demand falls as home solar peaks and then surges again in the evening), small errors in this estimate can make demand look more volatile than it really is. Treat it as directionally reliable, not to-the-megawatt precise.

4.3 Reference Trading Price

ReferenceTradingPrice-`{2023..2026}`.csv

Resolution: **30 min**

Unit: **\$/MWh**

Rows loaded: **51,168**

This is the wholesale price of electricity in WA — what it cost, on average, to buy one megawatt-hour (one full tank, in Section 2's terms) of power during each half-hour trading interval. It's not what households pay on their power bill (that's a separate, much smoother retail tariff); it's the raw market-clearing price paid to the reservoir operators, and, ultimately, flows through to retail prices over time.

WHY PRICES CAN GO NEGATIVE

A price below \$0/MWh means generators are effectively paying someone to take a tank of water off their hands. This isn't a data error — it happens in real markets when there's more low-cost water flowing in than the river needs (typically from wind and solar "wells"), and some generators would rather pay a small fee than shut their valve and restart it later. It's a genuine, if unusual, market outcome.

Administered price limits change over time

AEMO sets an official floor and ceiling that prices are not allowed to cross, and — importantly for comparing years — **those limits have moved**:

YEAR	MIN PRICE	TIMES HIT	MAX PRICE	TIMES HIT
2023 (Oct–Dec)	−\$716.71	2	\$738.00	43
2024	−\$1,000.00	44	\$738.00	123
2025	−\$150.52	1	\$1,100.00	1
2026 (to Aug)	−\$16.72	1	\$1,000.00	1

READING THIS TABLE CORRECTLY

In 2023–2024, the maximum price landed on *exactly* \$738.00 dozens of times — a sign the price was hitting a hard administered ceiling, not that the market happened to produce that number repeatedly. From 2025 onward the extreme prices are unique, one-off values well under the new higher ceiling — the market simply never got that stressed again. Comparing "price volatility" across years without knowing this makes 2024 look artificially wilder than 2026 on the downside, and masks how rare extreme prices became later.

Not every price is "real"

The file also carries a `Failure Reason` column. Most rows are blank (the price was set normally), but some intervals are flagged — the most important flag being `MarketAnalystOverride`, meaning a human manually set the price rather than letting the automated market engine calculate it (used on roughly 1.3–1.8% of intervals in 2025 and 2026). Any serious price analysis should be checked with these overridden intervals excluded.

4.4 Facility SCADA (Battery Storage)

FacilityScada-YYYY-MM.csv (filtered to battery facilities)

Resolution: **5 min**

Unit: **MWh → MW**

Rows loaded: **1,096,453**

DEFINITION

SCADA stands for *Supervisory Control and Data Acquisition* — the industrial monitoring systems that watch physical equipment (here, grid-scale batteries) in real time and report back what it's doing. In this guide's analogy, this file is the gauge mounted directly on each water tower: one row per tower per 5-minute check.

Two things make this dataset different from the others: the unit conversion covered in Section 2, and a sign convention that tells you which direction the tower is moving water.

SIGN	MEANING
Positive MW	Discharging — the tower is draining, sending water out to help supply the river
Negative MW	Charging — the tower is filling, drawing water from the river to store

A quirk worth knowing: "System Size (MW)" isn't the tower's fill or drain rate

You might expect a tower registered at, say, 300 MW to be able to fill at up to 300 MW and separately drain at up to 300 MW. It's actually recorded as the *sum* of both directions — the combined two-way pipe capacity, not a one-way rating:

FACILITY	MAX DISCHARGE	MAX CHARGE	OBSERVED SPAN	REGISTERED "SYSTEM SIZE"
KWINANA_ESR1	100.7 MW	−100.1 MW	200.8 MW	200 MW
COLLIE_ESR1	200.0 MW	−200.1 MW	400.1 MW	400 MW
COLLIE_BESS2	300.1 MW	−300.0 MW	600.0 MW	600 MW

WHY THIS MATTERS

Quoting `System Size (MW)` as "how big is this battery" overstates its real power rating by roughly 2×. `COLLIE_BESS2` is a **300 MW** tower — it can fill (charge) at up to 300 MW and, separately, drain (discharge) at up to 300 MW — not a 600 MW one. The registered figure is a two-sided span, not a one-way rating. This project's registry loader keeps the raw figure but this table should be consulted before quoting battery capacity anywhere.

4.5 Facility Registry

facilities.csv

Resolution: **snapshot**

Rows loaded: **176**

Every other dataset in this project refers to reservoirs, wells, towers, and large taps by a short **facility code** (e.g. `COLLIE_BESS2`, `ALCOA_KW_IL`). This file is the directory that turns those codes into something meaningful: who owns the facility, what class of participant it is, and (nominally) how big it is.

THE ONE THING THIS FILE IS MISSING

There is no "technology" or "fuel type" column — nothing directly says "this one is wind" or "this one is a water tower." This project infers it instead from a naming convention in the facility code's suffix: `_WF` for wind farm, `_PV` / `_SF` for solar, `_ESR` / `_BESS` for storage, `_GT` / `_U#` / `_G#` for thermal (gas/coal). This is a **stated assumption**, not something AEMO officially certifies — and it only successfully classifies 61 of the 176 registered facilities. The rest are mostly Demand Side Programme registrations (large industrial loads that get paid to reduce consumption on request) whose codes don't follow this pattern. It is reliable, however, for the seven water towers this project's battery analysis actually depends on.

4.6 Facility Maximum Daily Temperature

Resolution: **daily**

Unit: **°C**

facility-temperature-{2024..2026}.csv

This is the odd one out: everything else in the project is about the river, but this file is weather — the maximum temperature recorded near each facility on each day. It exists because temperature is one of the strongest drivers of electricity demand that has nothing to do with the market itself.

WHY IT'S HERE

Hot days push air-conditioning load up sharply — more taps running at once, in this analogy — which is exactly the kind of demand spike that can push wholesale prices toward their ceiling (Section 4.3) and lean hardest on the water towers (Section 4.4) to smooth things out. Joining this file against demand and price lets an analysis ask “is this price spike a heatwave story or something else?” instead of treating every spike as unexplained.

Resolution here is daily, not 5-minute or 30-minute — so this is a variable you join *onto* the finer-grained series (typically by matching date and, loosely, region), not one you analyse at the same granularity as the rest.

SECTION 05

How the six datasets connect

None of these files is interesting entirely on its own — the project's actual insights come from lining them up against each other. Here's the mental map, still in river terms:

DATASET	ROLE IN THE STORY
Distributed PV	Rain-barrel water that quietly reduces what the reservoirs have to put into the river, before anyone else in the chain even sees it.
Operational Demand & Withdrawal	What's left for the reservoirs to supply, split apart into "genuine tap use" vs. "towers topping themselves up."
Facility Temperature	The weather variable that explains a large share of <i>why</i> tap demand moves the way it does, day to day.
Facility SCADA (batteries)	The moment-by-moment record of tower filling (adds to demand) and draining (acts like extra supply), matched against the registry to know which tower is which.
Facility Registry	The directory that turns every facility code appearing elsewhere into an owner, a class, and an (inferred) technology.
Reference Trading Price	The financial outcome of all of the above: what it cost to keep the river's supply and demand balanced in each half-hour.

A typical question this project can answer sits at the intersection of several of these: "*On the hottest days, how much did the water towers refilling add to Operational Demand, and did that push the trading price toward its ceiling?*" — which touches temperature, demand/withdrawal, SCADA, and price all at once.

Reading the raw files without getting burned

These are file-format quirks, not conceptual ones — but they've each caused real bugs in this project's history, so they're worth knowing before you open a CSV directly instead of going through the loaders in `src/wa_data.py`.

1. Files are cut on UTC boundaries, but stamped in local time

`OperationalDemandWithdrawal-2025.csv` doesn't span the 2025 calendar year — it runs from 2025-01-01 08:00 to 2026-01-01 07:55 *local* time. The file is actually split at UTC midnight, which is 8am in Perth. Building a "2025 total" from that one file alone silently drops the first eight hours of January and picks up eight hours of the following January. Always concatenate all available files first, then slice by local calendar date — never trust one file to equal one year.

2. Rows arrive newest-first

The interval files list the most recent timestamp in row 1, oldest last — the opposite of what most people expect from a time series. The project's loaders re-sort everything into ascending order.

3. Dates are written day first

Interval files use `dd/MM/yyyy` (e.g. `01/09/2026` means 1 September, not 9 January) — the Australian convention, not the US `MM/dd/yyyy` one. `facilities.csv` and the temperature files use plain ISO dates (`2026-09-01`) instead. It's the same project, two different date formats depending on which file you're in.

4. "Extracted At" is not a revision log

Every file has an `Extracted At` column, but it's only ever filled in on the *first* row of the file — it's a single watermark saying "this whole file was downloaded at this moment," not a per-row record of when that particular reading was last revised or corrected.

5. Downloads can silently truncate

When many monthly battery SCADA files are downloaded in parallel, a network hiccup can cut a file off mid-line. If the cut happens to land exactly on a line boundary, the file looks like valid (just short) data rather than obviously broken — nothing will throw an error. This project's fetch script checks each file's final line against the expected pattern and re-downloads automatically if it looks truncated.

Glossary

Every term used in this guide, gathered in one place for quick lookup — starting with the analogy key from Section 1.

THE ANALOGY KEY

Grid (SWIS) = river network · AEMO = water authority · generator = reservoir/well · MW = flow rate · MWh = volume collected · battery = water tower · rooftop solar = private rain barrel · price = cost per bucket at the reservoir gate.

AEMO

Australian Energy Market Operator — the body that runs the electricity market and publishes this data. The water authority, in this guide's analogy.

AWST

Australian Western Standard Time, UTC+8. WA's only timezone — it never changes for daylight saving.

Behind-the-meter

Electricity generated and used on the customer's side of their meter (e.g. rooftop solar) — invisible to the grid operator's direct measurements, like a private rain barrel the water authority can't see into.

Dispatch interval

The 5-minute cycle AEMO uses to physically balance the grid — the finest time resolution in this project; the water authority's valve-check cycle.

Failure Reason

A flag on the price file noting when a trading interval's price wasn't set by the normal automated process (e.g. a manual override).

Grid-scale

A generator or battery large enough, and formally registered, to be directly dispatched and monitored by AEMO — as opposed to small, invisible rooftop systems.

MW (megawatt)

A unit of power — a rate of energy flow, measured at an instant. The pipe-gauge unit: how fast water is moving right now.

MWh (megawatt-hour)

A unit of energy — a total amount, accumulated over time. The tank unit: how much water collected over some stretch of time.

NEM

National Electricity Market — the interconnected grid and market covering the eastern Australian states. Not connected to WA's grid, or river, at all.

Operational Demand

Total sent-out power from grid-connected generation, net of rooftop solar, including network losses — the total flow the reservoirs must supply, towers included.

Operational Withdrawal

Power drawn from the grid by scheduled loads (in practice, battery charging) — water diverted into a tower to refill it. Always ≤ 0 in this project's files.

SCADA

Supervisory Control and Data Acquisition — real-time sensor/monitoring data from physical equipment, here grid-scale batteries. The gauge mounted on each water tower.

Spot price / Reference Trading Price

The wholesale market-clearing price of electricity, published every 30-minute trading interval, in \$/MWh — the cost per bucket at the reservoir gate.

SWIS

South West Interconnected System — the physical electricity grid covering WA's populated South West, operated as an electrical island with no link to any other grid, or river system, at all.

Trading interval

The 30-minute window over which electricity price is settled — six dispatch intervals bundled together, labelled by its start time. The billing window, in this guide's analogy.

Unscheduled Operational Demand

Operational Demand with battery charging subtracted back out — the closest available estimate of genuine, underlying tap demand.

WEM

Wholesale Electricity Market — the market AEMO runs for Western Australia, distinct from the NEM.

Sources: AEMO WA public market data (data.wa.aemo.com.au); WEM Registration Technical Guide v4.2; 2024 WEM Electricity Statement of Opportunities. All figures above are reproduced from this project's own data audit (`scripts/audit.py`), run 1 September 2026 against files downloaded the same day — see `docs/DATA_NOTES.md` for the full technical record. This edition adds the river/water-tower analogy used throughout; no figures were changed.